



COURSE DESCRIPTION FORM

INSTITUTION FAST School of Computing, National University of Computer and Emerging Sciences, Islamabad

PROGRAM(S) TO BE EVALUATED BSSE Spring 2026

Course Description

Course Code	AI-2002		
Course Title	Artificial Intelligence (SE)		
Credit Hours	3 + 1		
Prerequisites by Course(s) and Topics			
Assessment Instruments with Weights (homework, quizzes, midterms, final, programming assignments, lab work, etc.)	Project and assignments will cover the programming part of the course.		
	Breakdown of Course Work (Total):		
	Sessional Exam 1	1	15%
	Sessional Exam 2	1	15%
	Quizzes	4	12.5%
	Assignments	3	12.5%
	Project	1	10%
	Final	1	35%
Grading Policy	Absolute grading		
Policy about missed assessment items in the course	Retake of missed assessment items (other than midterms/ final exam) will not be held. For a missed midterm/ final exam, an exam retake/ pretake application along with necessary evidence are required to be submitted to the department secretary. The examination assessment and retake committee decides the exam retake/ pretake cases.		
Course Plagiarism Policy	Plagiarism in project or midterms/ final exam may result in F grade in the course. Plagiarism in an assignment will result in zero marks in the whole assignments category.		
Lab Instructors (if any)			
Course Coordinator	Dr. Muhammad Bilal		
URL (if any)			
Current Catalog Description	This course introduces students to the basic knowledge representation, problem solving, and learning methods of artificial intelligence. Upon completion, students should be able to develop intelligent systems by assembling solutions to concrete computational problems; understand the role of knowledge representation, problem solving, and learning in intelligent-system engineering; and appreciate the role of problem solving, vision, and language in understanding human intelligence from a computational perspective.		
Textbook (or Laboratory Manual	• Artificial Intelligence: A Modern Approach, 4 th ed. S. Russell and P. Norvig, Prentice Hall, 2020.		

for Laboratory Courses)	- Artificial Intelligence: Structures and Strategies for Complex Problem Solving, 6th ed. G. Luger, Addison Wesley, 2009 - AI Algorithms, Data Structures, and Idioms in Prolog, Lisp and Java, G. Luger and W. Stubblefield, Addison Wesley, 2009 - Artificial Intelligence: A Systems Approach. M. Tim Jones, Infinity Science Press, 2008																																										
Reference Material	Supporting website: http://aima.cs.berkeley.edu																																										
Course Learning Outcomes	<table border="1"> <tr> <th colspan="3">A. Course Learning Outcomes (CLOs)</th></tr> <tr> <td colspan="3">On successful completion of this course students will have to know how of:</td></tr> <tr> <td>1.</td><td>Understand the notions of rational behavior and intelligent agents.</td><td></td></tr> <tr> <td>2.</td><td>Develop a general appreciation of the goals, subareas, achievements and difficulties of AI.</td><td></td></tr> <tr> <td>3.</td><td>Knowledge of methods of blind as well as informed search and ability to practically apply the corresponding techniques.</td><td></td></tr> <tr> <td>4.</td><td>General understanding of major concepts and approaches in knowledge representation, planning, learning, robotics and other AI areas.</td><td></td></tr> <tr> <td>5.</td><td>Developing programming skills for AI applications.</td><td></td></tr> <tr> <th colspan="3">B. Program Learning Outcomes</th></tr> <tr> <td colspan="3">For each attribute below, indicate whether this attribute is covered in this course or not. Leave the cell blank if the enablement is little or non-existent.</td></tr> <tr> <td>1. Software Engineering Knowledge</td><td>To apply knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of Complex Software Engineering problems.</td><td>✓</td></tr> <tr> <td>2. Problem Analysis</td><td>Identify, formulate, research literature, and analyze complex computational problems, reaching substantiated conclusions using first principles of mathematics, natural sciences, computing, and software Engineering..</td><td>✓</td></tr> <tr> <td>3. Design/Develop Solutions</td><td>Design solutions for complex computing problems and design systems, components, and processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations..</td><td>✓</td></tr> <tr> <td>4. Investigation & Experimentation</td><td>Conduct investigation of complex computing problems using research-based knowledge and research based methods.</td><td>✓</td></tr> <tr> <td>5. Modern Tool Usage</td><td>Create, select, and apply appropriate techniques, resources and modern Computer-Aided software Engineering (CASE) tools, including prediction and</td><td>✓</td></tr> </table>	A. Course Learning Outcomes (CLOs)			On successful completion of this course students will have to know how of:			1.	Understand the notions of rational behavior and intelligent agents.		2.	Develop a general appreciation of the goals, subareas, achievements and difficulties of AI.		3.	Knowledge of methods of blind as well as informed search and ability to practically apply the corresponding techniques.		4.	General understanding of major concepts and approaches in knowledge representation, planning, learning, robotics and other AI areas.		5.	Developing programming skills for AI applications.		B. Program Learning Outcomes			For each attribute below, indicate whether this attribute is covered in this course or not. Leave the cell blank if the enablement is little or non-existent.			1. Software Engineering Knowledge	To apply knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of Complex Software Engineering problems.	✓	2. Problem Analysis	Identify, formulate, research literature, and analyze complex computational problems, reaching substantiated conclusions using first principles of mathematics, natural sciences, computing, and software Engineering..	✓	3. Design/Develop Solutions	Design solutions for complex computing problems and design systems, components, and processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations..	✓	4. Investigation & Experimentation	Conduct investigation of complex computing problems using research-based knowledge and research based methods.	✓	5. Modern Tool Usage	Create, select, and apply appropriate techniques, resources and modern Computer-Aided software Engineering (CASE) tools, including prediction and	✓
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		modelling for complex computing problems.	
	6. Society Responsibility	Understand and assess societal, health, safety, legal, and cultural issues within local and global contexts, and the consequential responsibilities relevant to professional computing practice	
	7. Environment and Sustainability	Understand the impact of professional software solutions in societal and environmental contexts and demonstrate knowledge of, and need for, sustainable development.	
	8. Ethics	Apply ethical principles and commit to professional ethics and responsibilities and norms of Software Engineering practice.	✓
	9. Individual and Team Work	Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.	✓
	10. Communication	Communicate effectively on complex Software Engineering processes and activities with the software Engineering community and with society at large.	
	11. Project Management and Finance	Demonstrate knowledge and understanding of management principles and economic decision-making and apply these to one's own work as a member or a team.	
	12. Life Long Learning	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological changes.	✓

C. Mapping of CLOs to PLOs (CLO: Course Learning Outcome, PLOs: Program Learning Outcomes)													
		PLOs											
		1	2	3	4	5	6	7	8	9	10	11	12
CLOs	1	✓	✓										✓
	2	✓	✓										
	3	✓	✓	✓	✓								
	4				✓	✓				✓			
	5	✓	✓	✓		✓			✓	✓			✓

Topics Covered in the Course, with Number of Lectures on Each Topic (assume 15-week instruction and 3 contact hours per week)	Topics to be covered:			
	List of Topics	No. of Weeks	Contact Hours	CLO(s)
	Introduction to AI	1	3	1
	Intelligent Agents: Problem Solving, Agent types, Agent environments	1	3	2,3,5
	Search problem formulation, Search space definition, Tree Search, Graph Search, Uninformed searches - BFS, DFS, UCS	1	3	2,3,4,5
	Informed Search: Heuristic Function, Greedy Best First Search, and A* search. Problems with Search	1	3	2,3,4,5
	Local Search: Hill Climbing, Beam search, simulated annealing	1	3	2,3,4,5
	Optimization: Genetic Algorithm.	1	3	2,3,4,5
	Constraint Satisfaction Problems Backtracking search for CSPs, Problem structure and problem decomposition, Local search for CSPs, Cryptarithmic Problem. Forward checking, AC3, MRV, LCV	2	6	2,3,4,5
	Game as Search Problems, Adversarial search Minimax and alpha beta pruning.	1	3	2,3,4
	Intro to Machine Learning, Supervised Learning, KNN, Decision Tree	1	3	2,3,4
	Unsupervised Learning, K-Means, Hierarchical Clustering	1	3	2,3,4
	Neural Networks: Perceptron, Multilayer perceptron, Applications of neural networks, Perceptron Learning, Back-propagation, Gradient Descent	3	9	1,4,5
	Intro to Reinforcement Learning, Q-Learning	1	3	2,3,4
	Total	15	45	
Laboratory Projects/Experiments Done in the Course	Lab Manuals and Tasks			
Programming Assignments Done in the Course	Yes, All the assignments were programming based and the language should be python.			



Class Time Spent on (in percentage)	Theory (%)	Problem Analysis (%)	Solution Design (%)	Social and Ethical Issues (%)
	50	25	20	5
Oral and Written Communications	Every student is required to submit at least __4__ written reports of typically __4-5__ pages and to make __1__ oral presentation of typically __20__ minute's duration.			

Lab/ Practical Component of the course

Not applicable.